device authentication. These vary from different pre-shared keys to validating the client side TLS certificates. Kaa also supports Flexible Credentials Lifecycle Management. This credential management system allows us to provision, suspend or unsuspend, as well as revoke different credentials that are being used by various connected devices.

The advantages

Kaa has a number of critical advantages when it comes to rapid IoT development and deployment. Let's have a look at some of them.

1. ***Customisable microservices architecture*:** As we have already discussed, Kaa has a customisable architecture that fragments the different features of Kaa (like visualisation, messaging, etc) as microservices. This leads to easy customisation and ensures effective separation between different parts of the platform. All these Kaa microservices can be easily integrated with third party systems and can also be rearranged as per individual needs.
2. ***Freedom of choice regarding technology:*** Kaa facilitates usage of our favourite DevOps tools while implementing Kaa based solutions in any environment. This is possible because Kaa runs all its microservices in Docker containers and they use well defined REST based interfaces. It allows for the development and deployment of different applications virtually written in any programming language.
3. ***Scalable and self-healing:*** Kaa is very scalable by design and can be easily expanded by adding more servers to the Kaa cluster. It can reliably support as many endpoints as needed. It has got its own self-healing capabilities that can even restore the platform in case of any failures.
4. ***Freedom of deployment:*** Users have all the freedom to deploy different Kaa services anywhere they prefer. It can be deployed at one's own data centre or can be hosted on any public cloud infrastructure as well. It is really perfect for hybrid deployments.
5. ***Open IoT protocols*:** The Kaa platform supports messaging with connected devices using MQTT and CoAP. Both of these are widely used IoT protocols having wide implementations for all the mainstream languages.
6. ***Gateway support*:** Based on the type of device, one can choose among the two available connectivity options — either connect them directly or use an IoT gateway. IoT gateways are handy when the devices don't support any of the available Kaa protocols, as they can convert the data semantics as well as the contents of the device's messages into the protocols that are supported by the Kaa platform.
7. ***Application versioning:*** Kaa supports flexible application versioning. Different versions of applications allow us to define various functions and capabilities for our endpoints. For instance, as the IoT solution evolves over time, a couple of new capabilities can be introduced, while others get retired. The Kaa platform always maintains support for all of the application versions that are used by the devices in the field.
8. ***Multi-applications*:** Kaa helps organise the device ecosystem by isolating the device management functionality and data flows in the platform. One Kaa solution can support multiple Kaa applications, allowing for cross-application interaction.
9. ***Blueprints:*** Kaa provides a human-readable and easily understandable blueprint that fully describes the system functionality by defining the necessary microservices along with their configurations. Blueprints empower us with the code approach for managing platform clusters. A blueprint can be easily used to reliably deploy different identical instances of the same solution to various environments—test, development, staging, production, etc.
10. ***Built on open components:*** The Kaa platform basically relies upon a variety of popular and well-tested open source components. This helps flatten the learning curve for any new user and helps Kaa to be open and customisable from its very core.
11. ***Continuous support from vendors*:** The Kaa IoT platform is supported by the Kaa IoT company. Kaa vendor services include professional as well as engineering assistance to client teams and production support. END

Areas where Kaa can be used

Kaa is being widely used across all industries. But there are some among them which use Kaa to a greater extent for different IoT based applications. Here are a few of those verticals:

- Consumer electronics
- Automotive
- Healthcare
- Smart cities
- Telecommunications
- Wearables
- Logistics
- Agriculture
- Sports and fitness

References

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